

Mapping Best Empirical Models

A Meta-Analysis of Winning Solutions in Kaggle Tabular Competitions

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Motivation

Winning a tabular Kaggle competition is supposed to be about **elaborate feature engineering**.

The clever, hand-crafted features that separate the best from the rest.

But the winners disagree

Across the 45 winning solutions:

2

median own features
engineered
(mean 3.9)

6

winners engineered *none* at
all

62%

simply ensembled existing
models

Winning isn't the feature-engineering craft the folklore promises.

What we “know” is anecdotal

- Folk wisdom lives in blog posts, forum threads, and winner interviews.
- Academic benchmarks compare **algorithms** on fixed datasets, never **winners** against strong non-winners.
- Nobody has systematically asked what *distinguishes* a winning solution from a near-winning one.

This study codes 45 winning solutions + 11 near-winning controls.

Research questions

- RQ1** What strategy **paradigms** define winning? Do folklore “constraint → strategy” rules hold?
- RQ2** Which feature-engineering techniques do winners use, and how much?
- RQ3** Does feature engineering **distinguish** winners from strong non-winners?
- RQ4** How is the winning **population** structured?

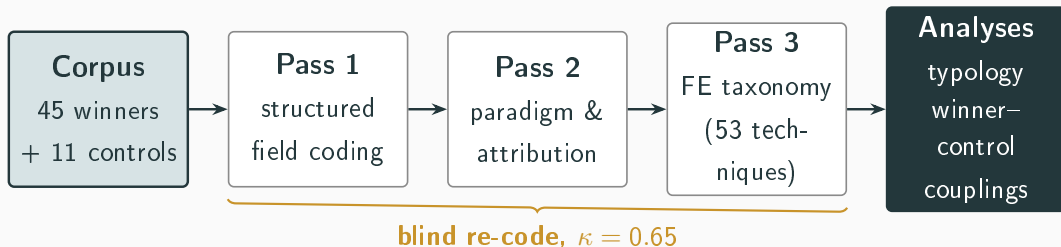
Data & Method

The corpus

- 45 winning solutions (top-three finishers)
- 11 near-winning controls (ranks 4–25, same competitions)
- February 2022 – April 2026
- Mostly Kaggle Playground Series (synthetic data); one real-data Featured competition
- Coded on a 40-column schema plus a 53-technique feature-engineering taxonomy
- Caveat: only solutions with public write-ups can be coded (documentation self-selection)

A content-analysis meta-study

No model is trained: the unit of analysis is **one completed solution**, coded in three passes.



Each winner is paired against a near-winning control from the *same* competition, isolating what distinguishes winning from merely strong play.

Checking our own coding

- Primary coding was *AI-assisted, researcher-directed*.
- A *blind re-code* of a stratified 12-entry subsample gives $\kappa = 0.65$: substantial agreement at the technique-family level.
- So FE results are reported at the *family* level, where they are reliable.

Limitations: small n (11 paired competitions); synthetic-data corpus; intra-rater (not inter-rater) reliability; documentation self-selection.

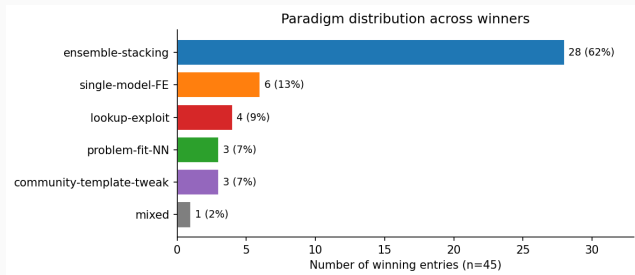
Findings

What winners actually use



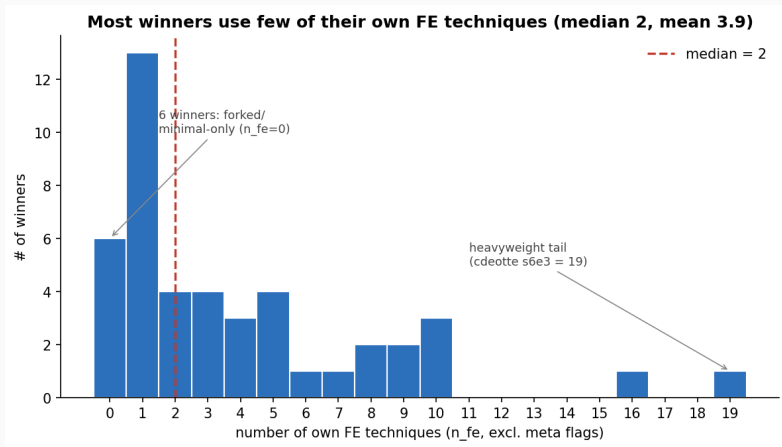
A few models (GBMs) and ensemble stacking dominate; feature engineering is a long,

Winning reduces to a few paradigms



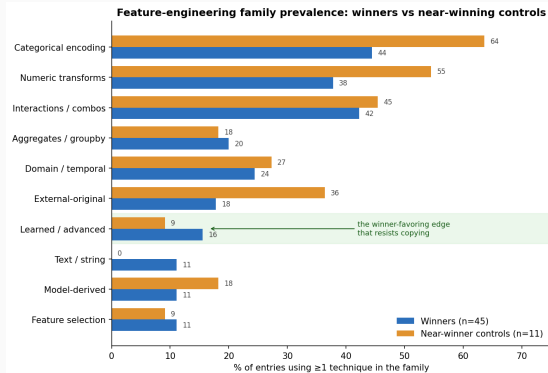
- **Ensemble stacking** = 62% of wins (28/45)
- Gradient-boosted trees are the dominant model in **35 of 45**
- A short tail: single-model-FE, lookup-exploit, problem-fit-NN

Winning FE is a small shared core



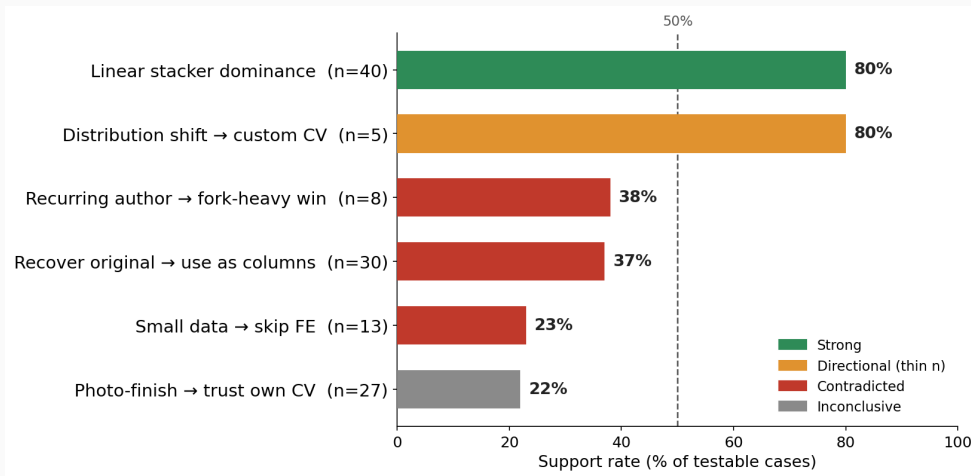
The median winner engineers 2 of its own features, and 6 used none. Target encoding is the one near-universal technique; the rest is a long tail.

FE does not separate winners from near-winners



- Aggregate: controls do *more* FE, but that's a confound (their competitions are FE-heavy)
- Paired, within-competition: winners do more in 7 of 11 (n.s.)
- **Learned features** are the one family that consistently leans winner

Folklore “rules” mostly don’t hold

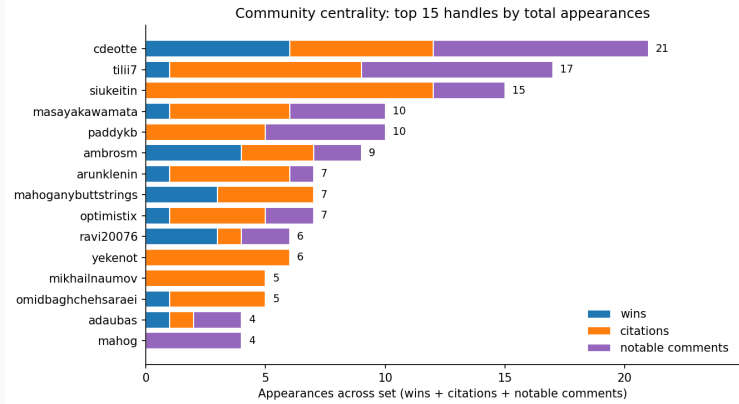


The lone green survivor, **linear-stacker dominance**, just restates standard ensembling. 12/16
The 80% “custom CV” rule rests on only **5 cases**.

The documented top is a recurring guild

45%

author overlap between
winners and near-winners



A few central figures supply widely-used techniques without always winning (originators vs. canonizers); recognition accrues to the already-prominent.

Conclusions

Three takeaways

1. Winning $\neq$ FE folklore: a small shared core, not elaborate craft.
2. Learned features are the one feature-level edge that resists copying from a public notebook.
3. Winning is, in part, a community phenomenon.

What this does, and does not, establish

- Claims are scoped to the **documented competitive elite**: not typical competitors, not real-world tabular ML.
- The corpus is mostly **synthetic** Playground data; some strategies are generator artifacts.
- The winner-control test is **underpowered** ($n = 11$): the result is “no clean separation,” not “no difference.”

Contributions & future work

Contributions

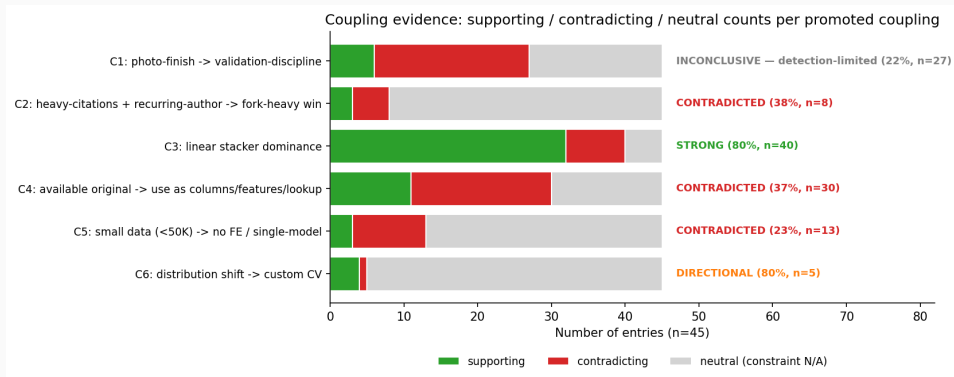
- The first structured, reliability-checked characterization of winners *versus near-winners*.
- A reusable 53-technique FE taxonomy and multi-pass coding workflow.
- Replaces folklore with documented frequencies.

Future work

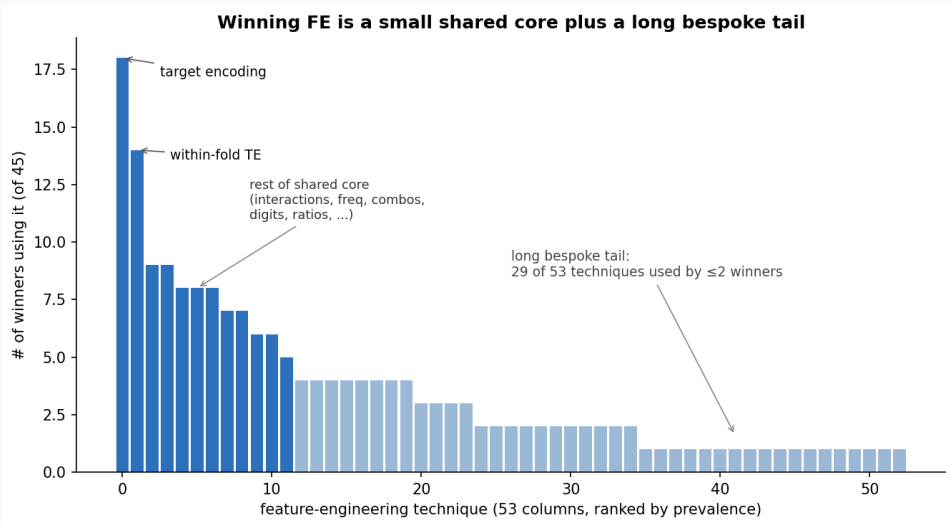
- Real-data Featured competitions; other platforms and modalities.

Winning turns less on the *volume* of feature engineering
than on **paradigm fit** and **who competes**.

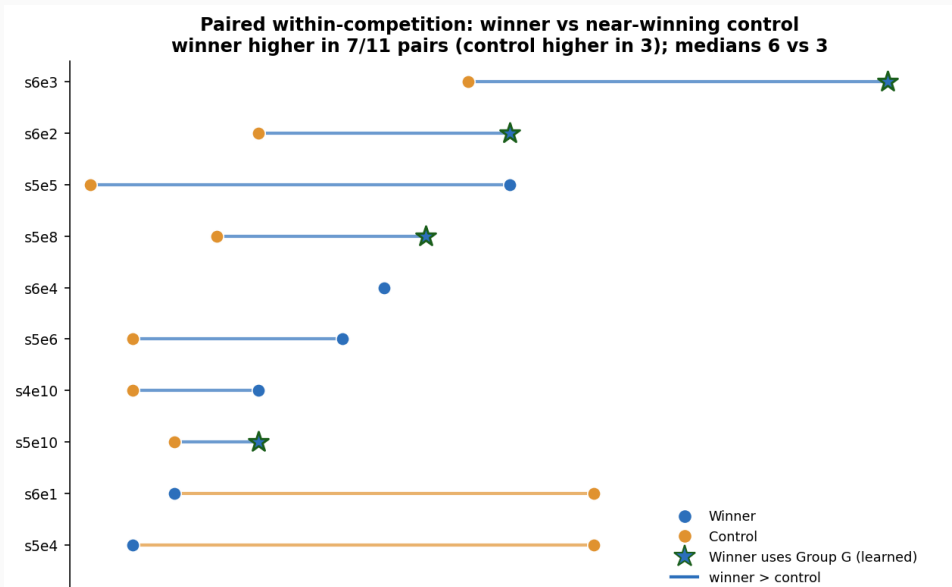
Appendix: coupling evidence (detailed)



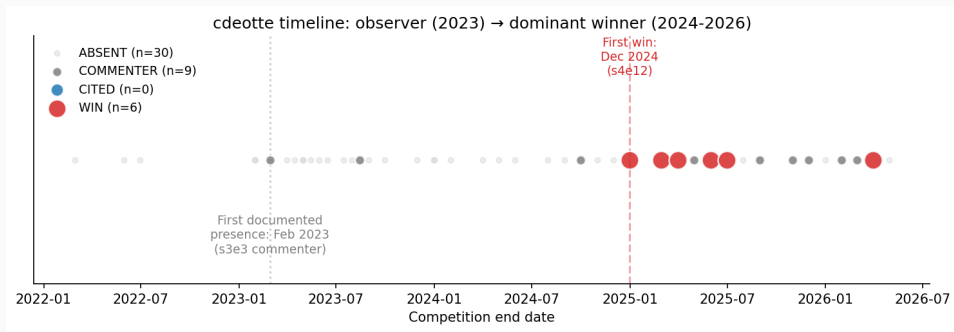
Appendix: feature-engineering technique long tail



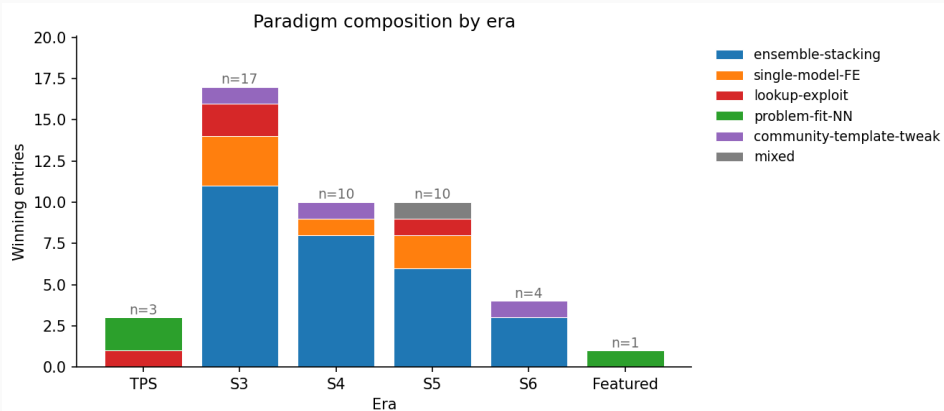
Appendix: paired winner vs. control (n_{fe})



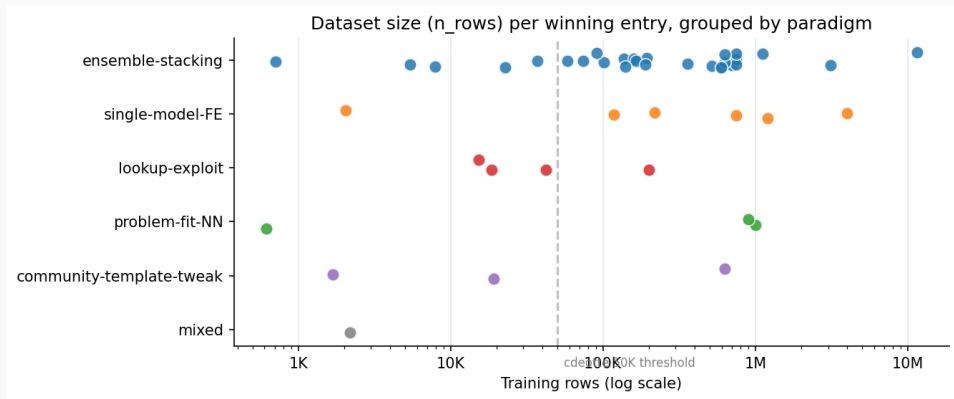
Appendix: cdeotte, observer to dominant winner



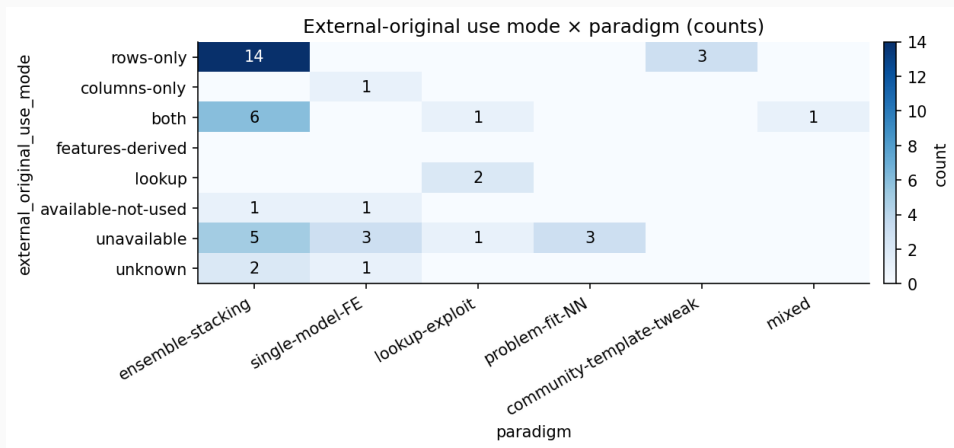
Appendix: paradigm composition by era



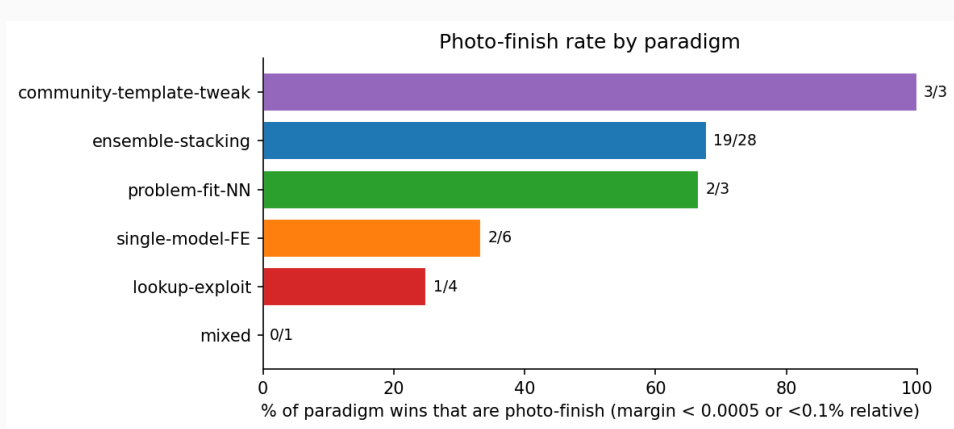
Appendix: dataset size by paradigm



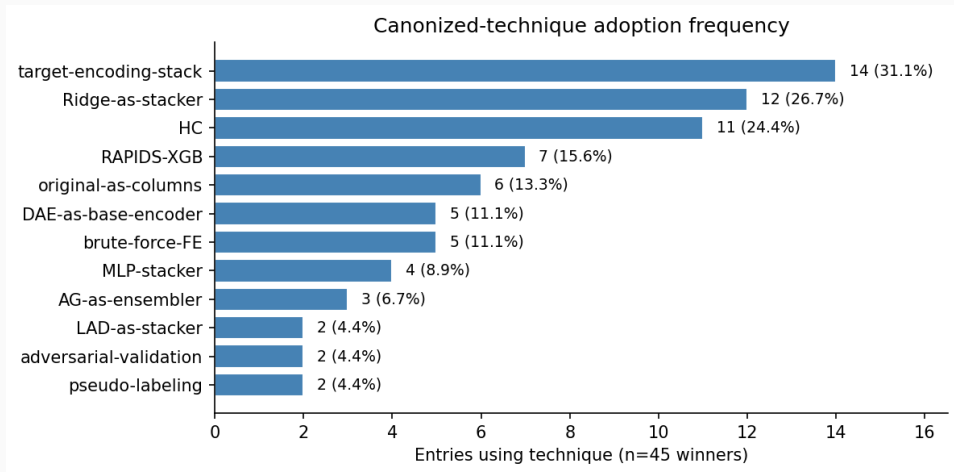
Appendix: external-original use by paradigm



Appendix: photo-finish rate by paradigm



Appendix: most-canonized techniques



Appendix: citations by origination score

